Electric Vehicle Data Quality Analysis Using Tableau Prep and Python

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Introduction

- Data quality is important for accurate analysis.
- Poor-quality data leads to incorrect insights.
- Tableau Prep helps clean and transform data.
- Python helps automate advanced validation.

Dataset Description

- Dataset: Electric Vehicle Population Dataset
- Contains EV manufacturer, model, range, location, and pricing information.
- Original dataset size: ~113,000 rows.
- Fields include Make, Model, Electric Range, State, Vehicle Location, MSRP, etc.

Problem Statement

- The Electric Vehicle dataset contained duplicate, missing, and inconsistent data.
- These issues affected data quality and analysis accuracy.
- Tableau Prep and Python were used to clean and transform the dataset.

Project Objectives

- Identify data quality issues
- Apply Tableau Prep transformations
- Integrate Python functions using TabPy
- Improve dataset consistency and accuracy
- Generate insights from cleaned data

Identified Data Issues

- Identified Problems:
- Duplicate rows
- Missing values
- Invalid Electric Range values
- Base MSRP = 0
- Inconsistent state names
- Mixed coordinate formatting
- Unstructured location data

Tableau Prep Workflow

Workflow Steps:

1. Input Step
2. Clean Step
3. Duplicate Removal
4. Filtering Invalid Values
5. Calculated Fields
6. Python Integration
7. Aggregate Step
8. Output Step

Transformation Steps

Transformations Applied:

1. Duplicate Removal
2. Filter Electric Range
3. Filter Base MSRP
4. Standardize Make Names
5. Create State Full Name
6. Create Vehicle Age
7. Convert Datatypes
8. Remove Special Characters
9. Split Coordinates
10. Rename Fields
11. Hide Unnecessary Fields
12. Aggregate Data

Duplicate Removal

Problem:

Duplicate vehicle records existed in the dataset.

Solution:

- Used ‘Identify Duplicate Rows’
- Filtered to keep only unique records

Filtering Invalid Values

Invalid Values Removed:

- Electric Range = 0
- Base MSRP = 0

Reason:

These values were considered invalid for analysis.

Calculated Fields

Calculated Field 1:

State Full Name

Example:

- WA → Washington
- OR → Oregon

Calculated Field 2:

Vehicle Age

Formula: 2026 - Model Year

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Coordinate Splitting

Problem:

Vehicle coordinates were stored in combined format.

Solution:

- Removed unnecessary characters
- Split longitude and latitude values

Python Integration

Python Integration:

- Used TabPy
- Connected Tableau Prep to Python
- Used Pandas library

Python Functions:

1. Missing Value Detection
2. Outlier Detection

Python Function 1

- Missing Value Detection
- Purpose: Detect rows containing null values.
- Output field: Missing_Check
- Python was used to automate missing-valueanalysis.

Python Function 2

- Outlier Detection
- Purpose: Identify abnormal Electric Rangevalues.
- Output field: Outlier_Flag
- Statistical logic using mean and standarddeviation was applied.

Aggregate Step

Aggregation:

Grouped data by manufacturer.

Result:

Calculated EV count by manufacturer.

Final Cleaned Dataset

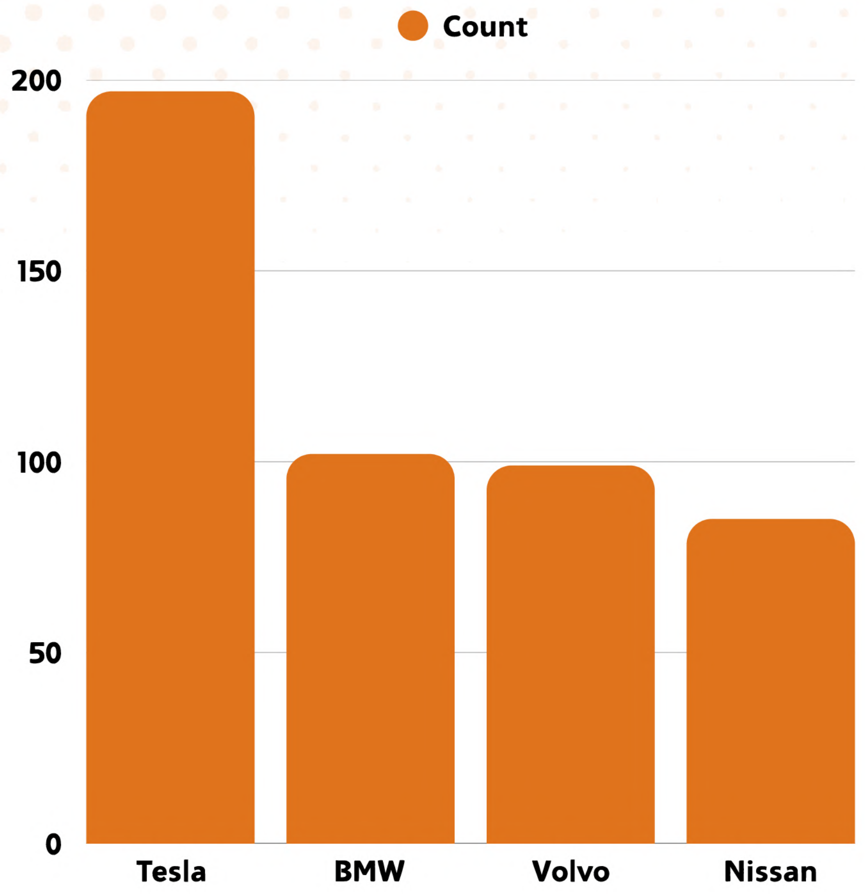
Improvements Made:

- Duplicate records removed
- Invalid values filtered
- Coordinates separated
- State names standardized
- Vehicle age calculated
- Data consistency improved

Insights and Findings

Key Insights:

- Tesla had the highest EV registrations
- Washington had the highest EV population
- Most EV ranges were normal
- Data quality improved significantly



Tesla had highest EV registrations

Conclusion

- Successfully cleaned EV dataset
- Applied 10+ Tableau Prep transformations
- Integrated Python automation
- Improved data quality and consistency
- Learned practical data-cleaning skills

Thank You